

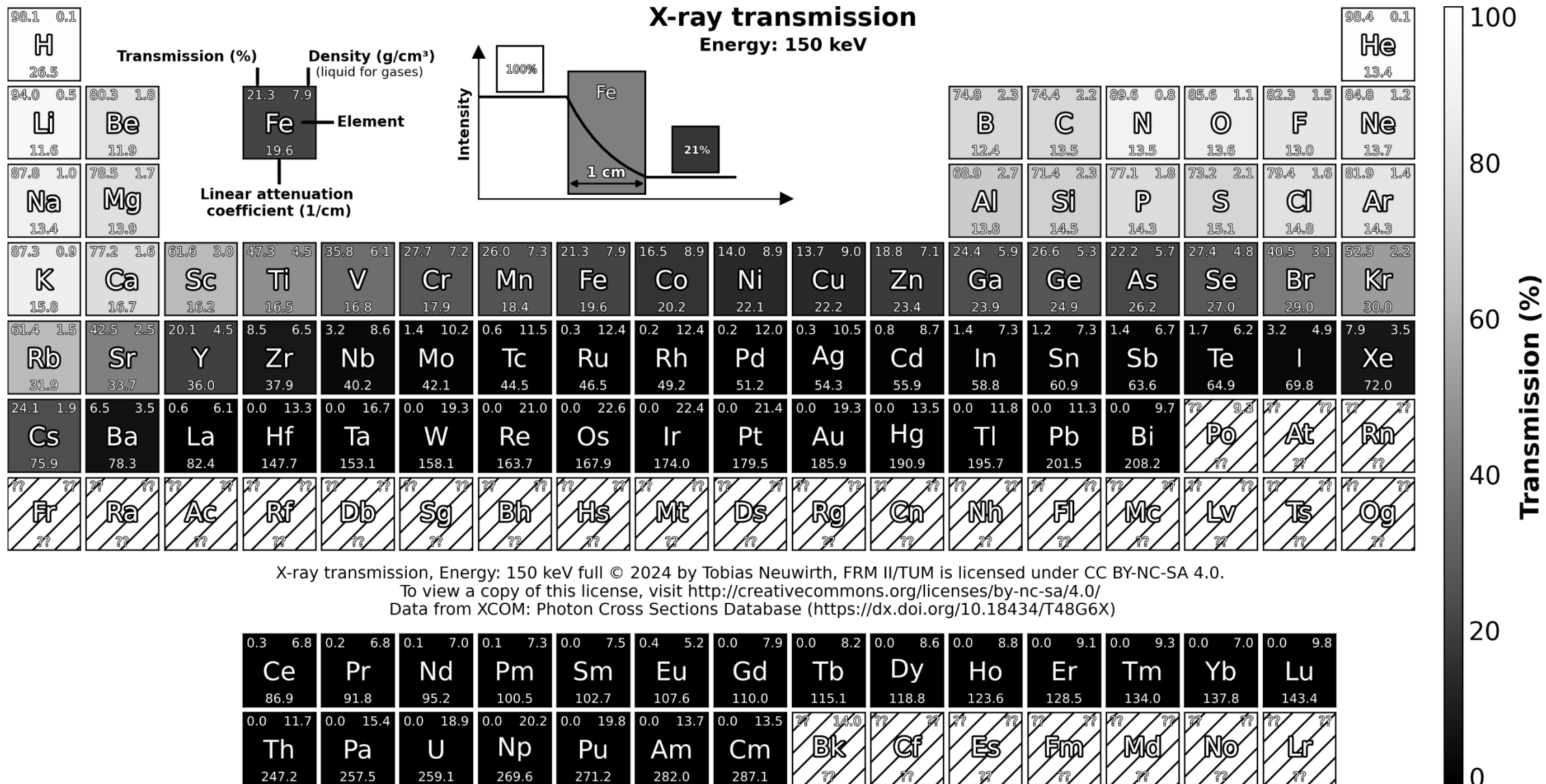


Experimental methods in solid-state physics

06 Neutron Scattering and Mössbauer Spectroscopy

Dr. Christoph Kastl
Walter Schottky Institut and Physics Department
Zentrum für Nanotechnologie
und Nanomaterialien

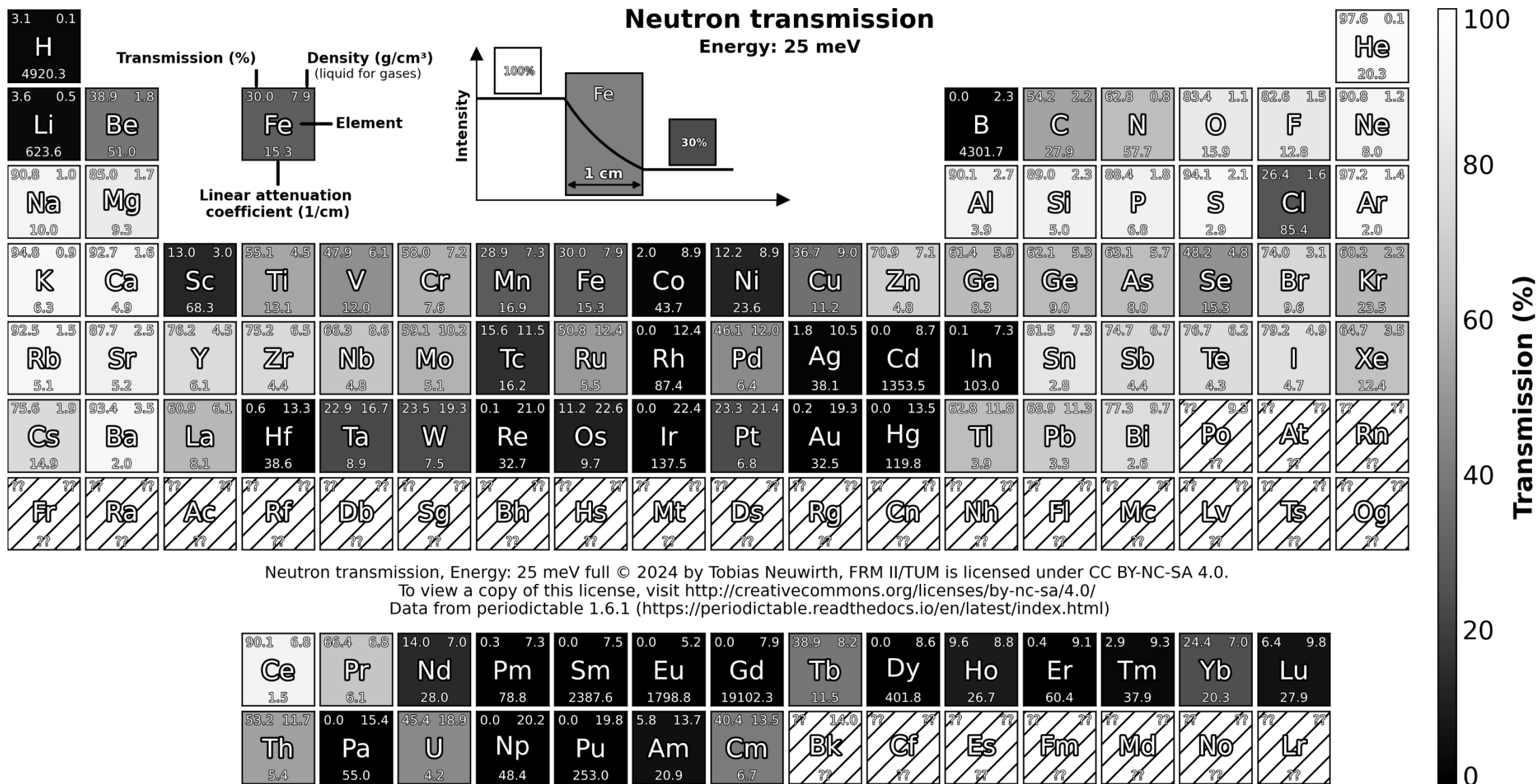
Attenuation of X-rays in matter



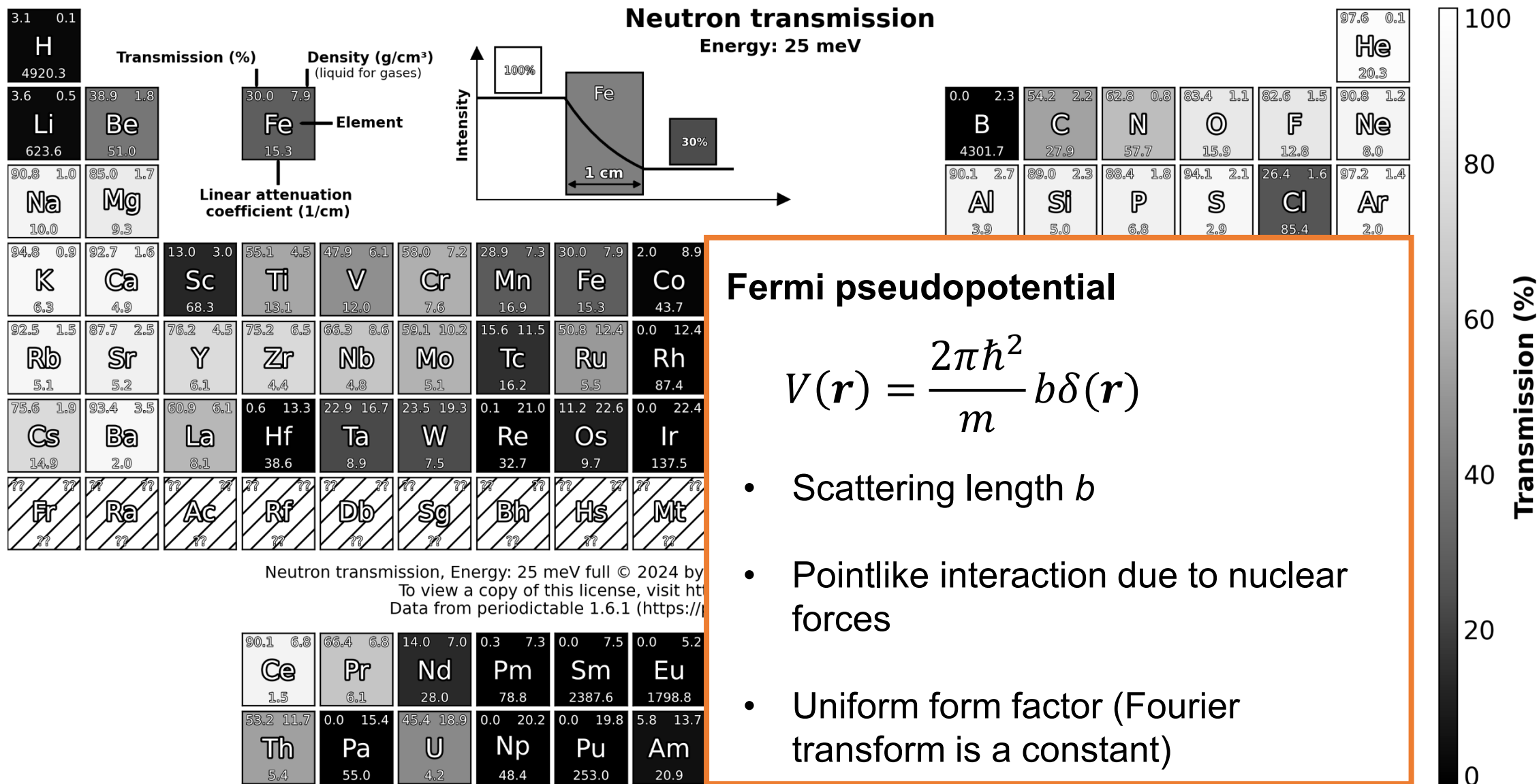
FRM – II Garching



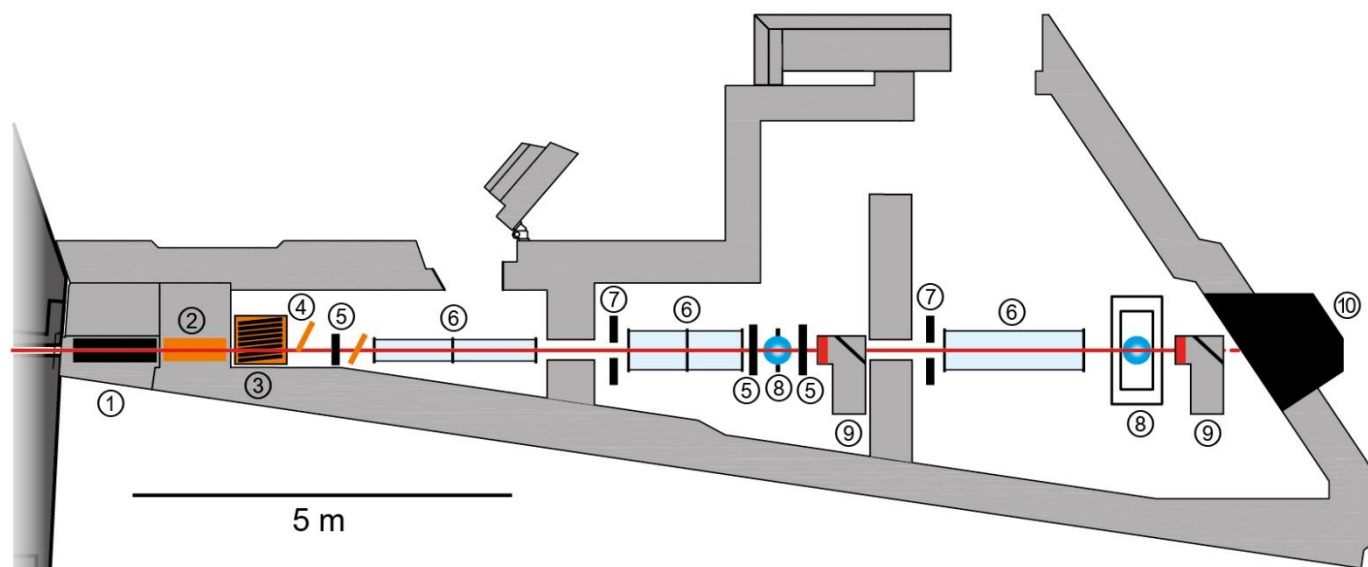
Attenuation of thermal neutrons



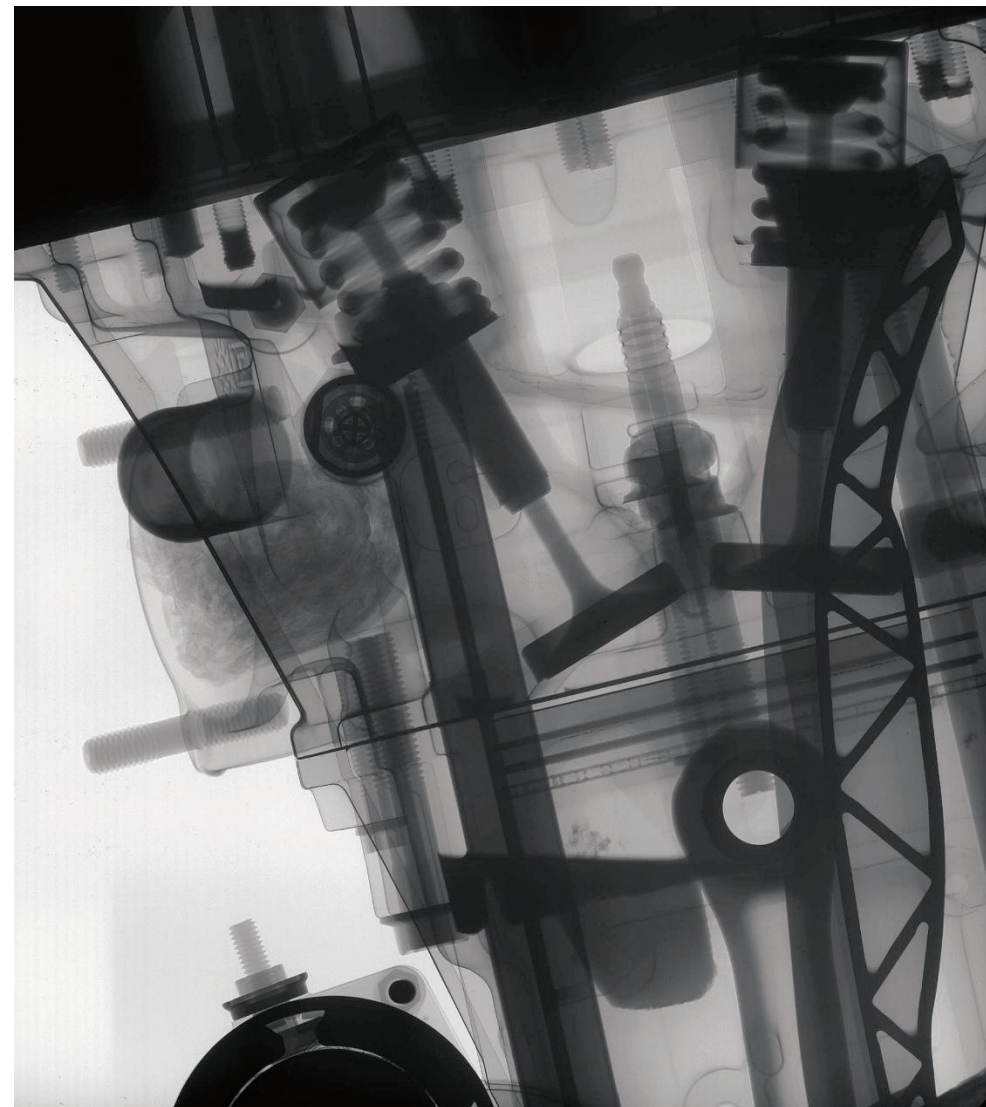
Attenuation of thermal neutrons



Antares beamline at FRM - II



- | | |
|---|-------------------------|
| ① Beam shutter | ⑥ Flight tubes |
| ② Collimator | ⑦ Slits |
| ③ Velocity selector | ⑧ Sample position 1+2 |
| ④ Double crystal monochromator (optional) | ⑨ Detector position 1+2 |
| ⑤ Neutron grating interferometer (optional) | ⑩ Beam stop |



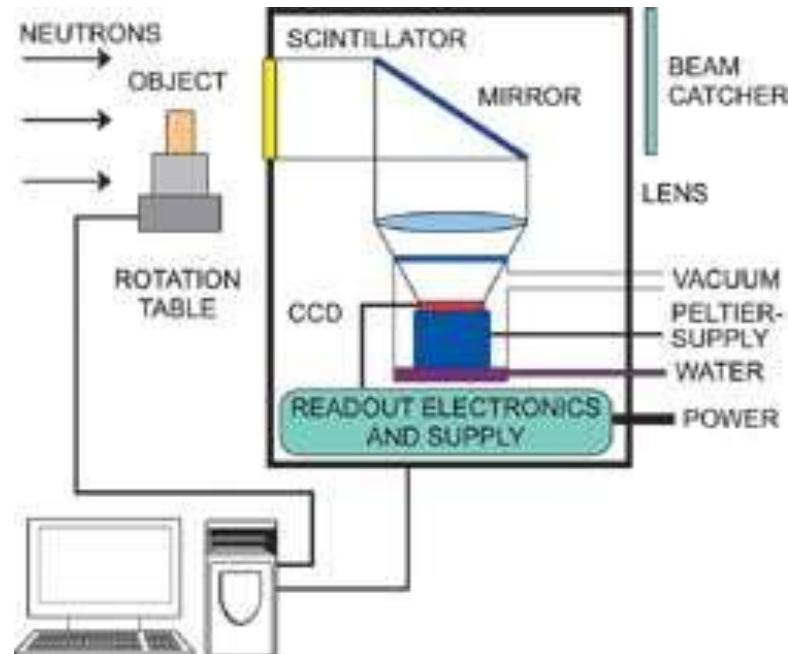


Detection of thermal or cold neutrons



^3He tube

- Neutron capture $^3\text{He} \rightarrow ^3\text{H} + ^1\text{H}$ (764 keV)
- Impact ionization in the gas tube $\rightarrow$ current pulse
- Gamma rays interact weakly with $^3\text{He} \rightarrow$ excellent discrimination of neutron events vs. background counts

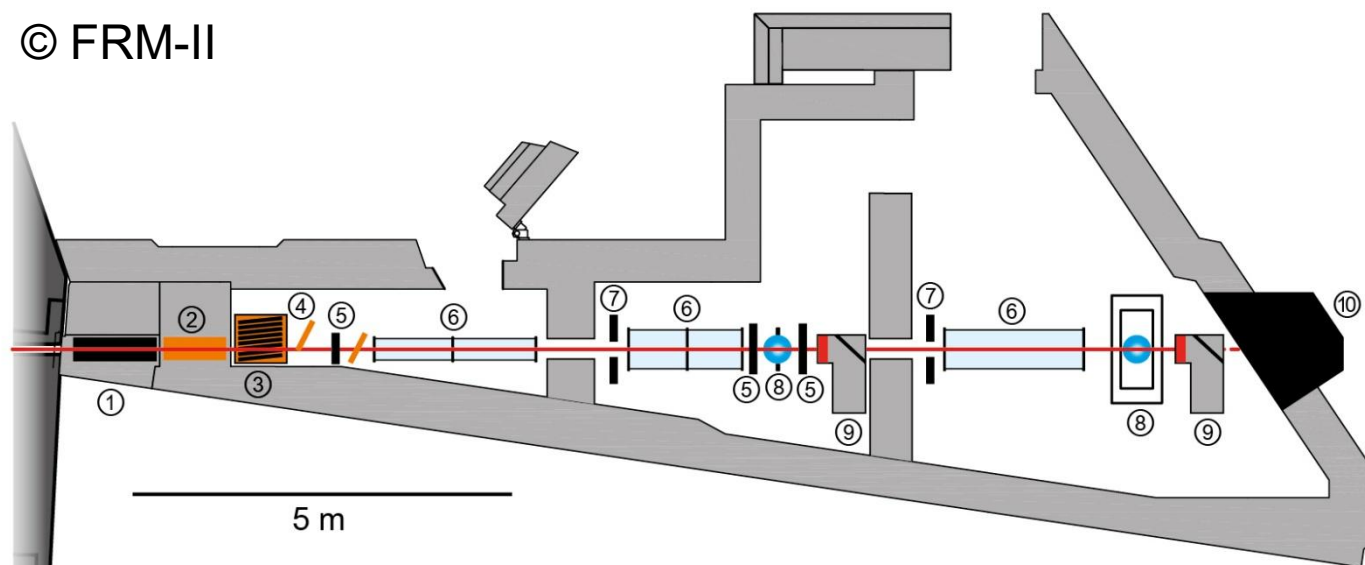


^6Li scintillator

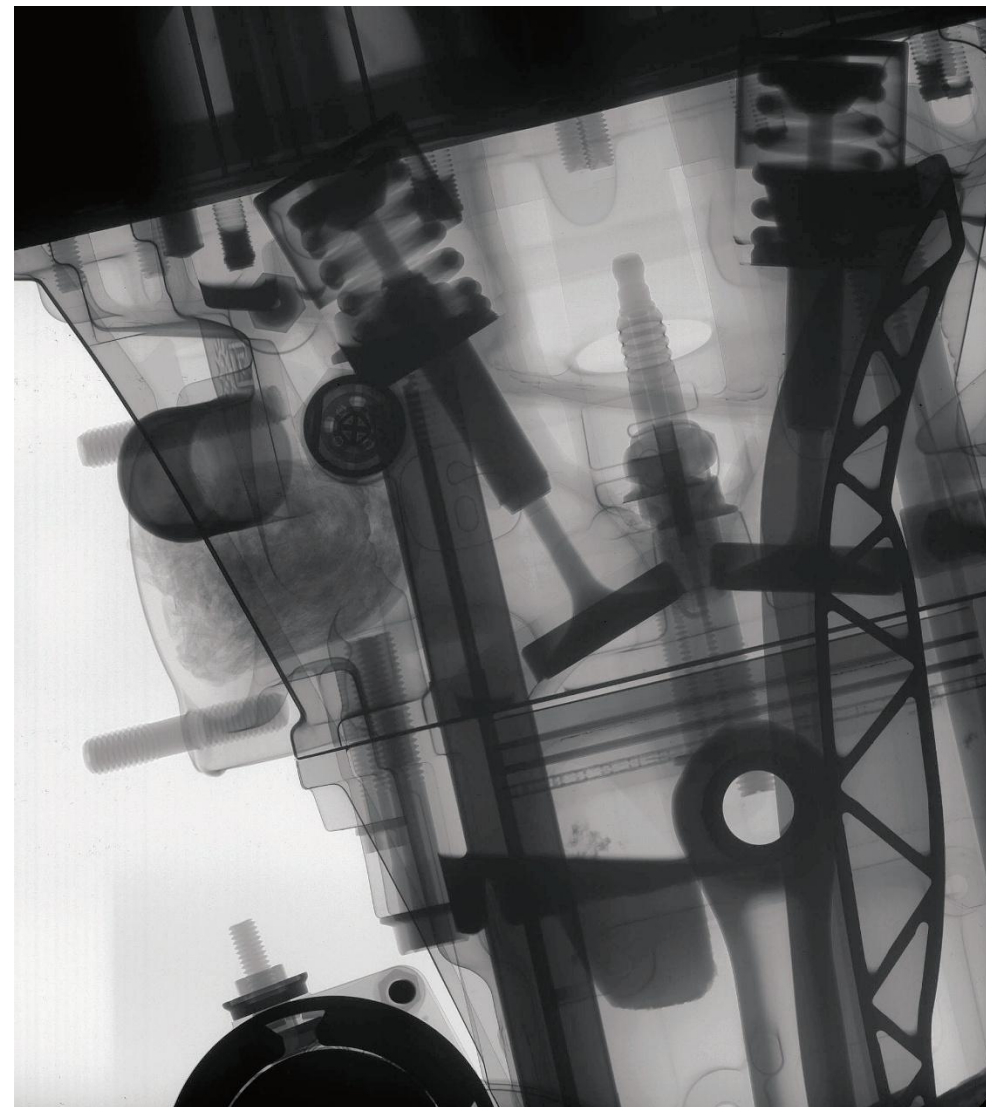
- Glass material enriched with ^6Li
- Neutron capture $^6\text{Li} \rightarrow ^3\text{H} + \alpha^1$ (4.78 MeV)
- Reaction products have limited range in the Li-containing material ($\sim 10 \mu\text{m}$)
- Conversion of the reaction energy into UV/visible scintillation light $\rightarrow$ detection with regular CCD camera (spatial resolution) or photomultiplier tube (energy resolution).

Antares beamline at FRM - II

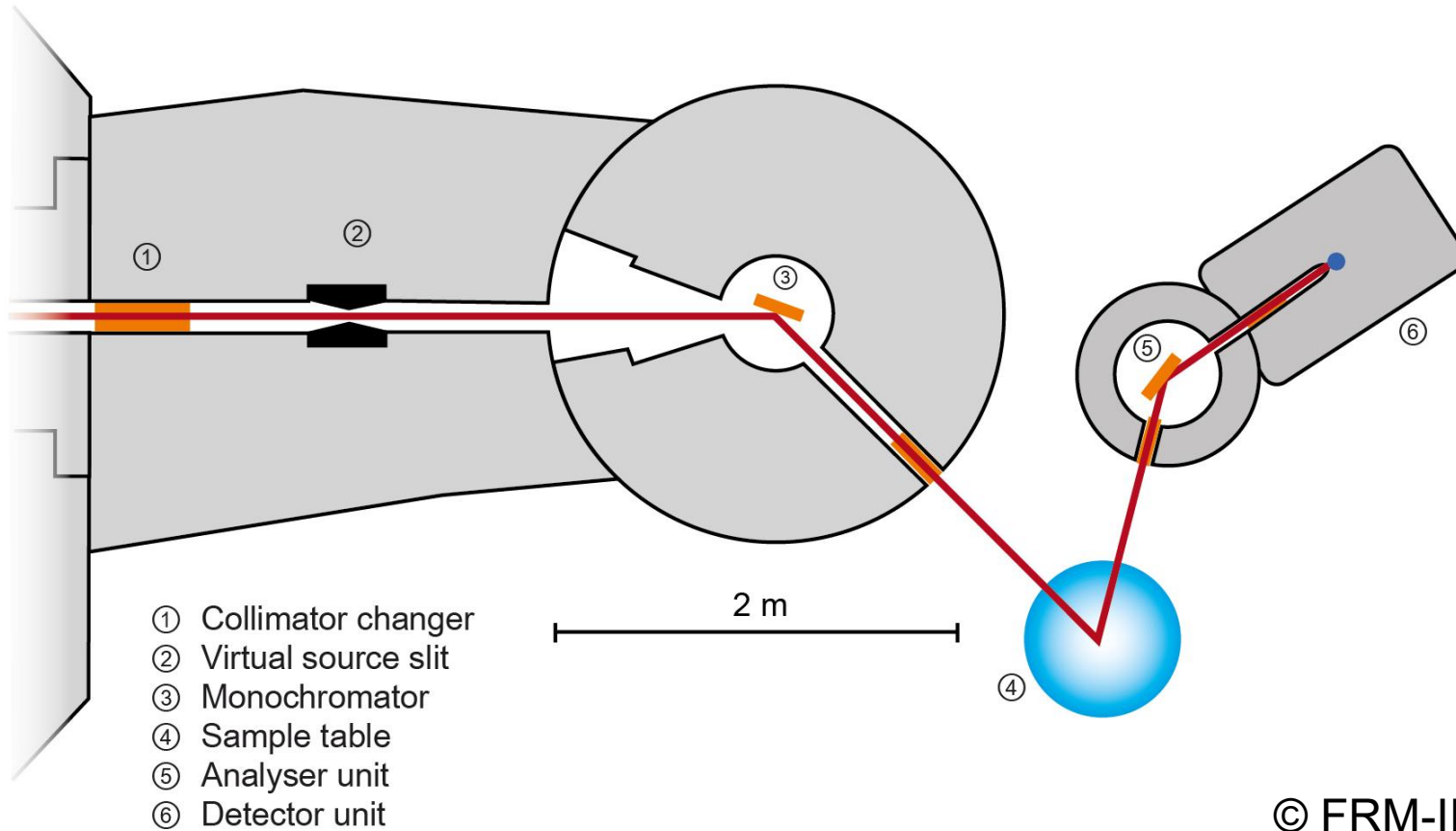
© FRM-II



- | | |
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Triple axes spectrometer

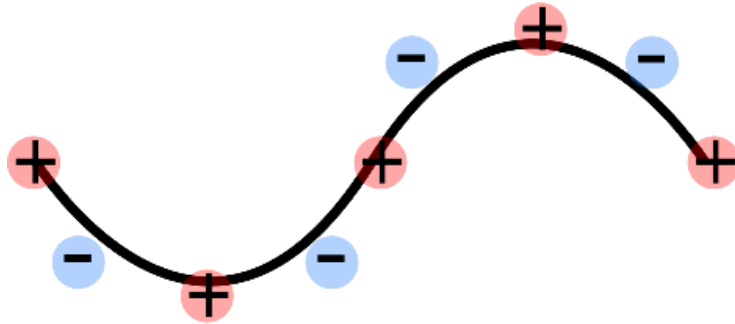


Inelastic scattering

$$q = k - k'$$

$$\Delta E = \hbar\omega = E - E'$$

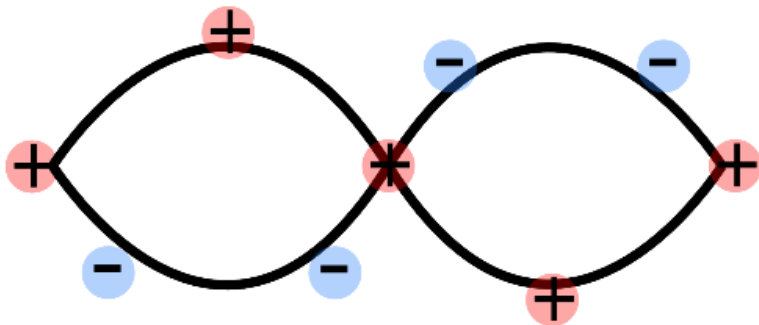
Acoustical Mode $\rightarrow$ sound wave



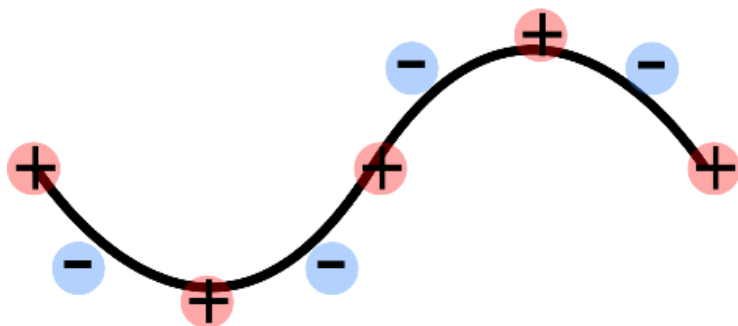
Number of **phonon branches**: $3N$

N : number of atoms in the primitive unit cell

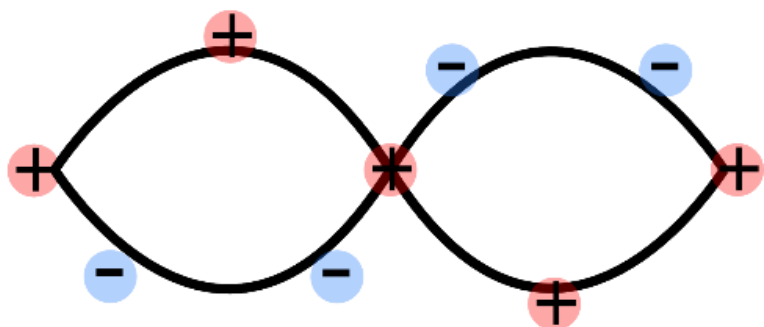
Optical Mode $\rightarrow$ interacts with light



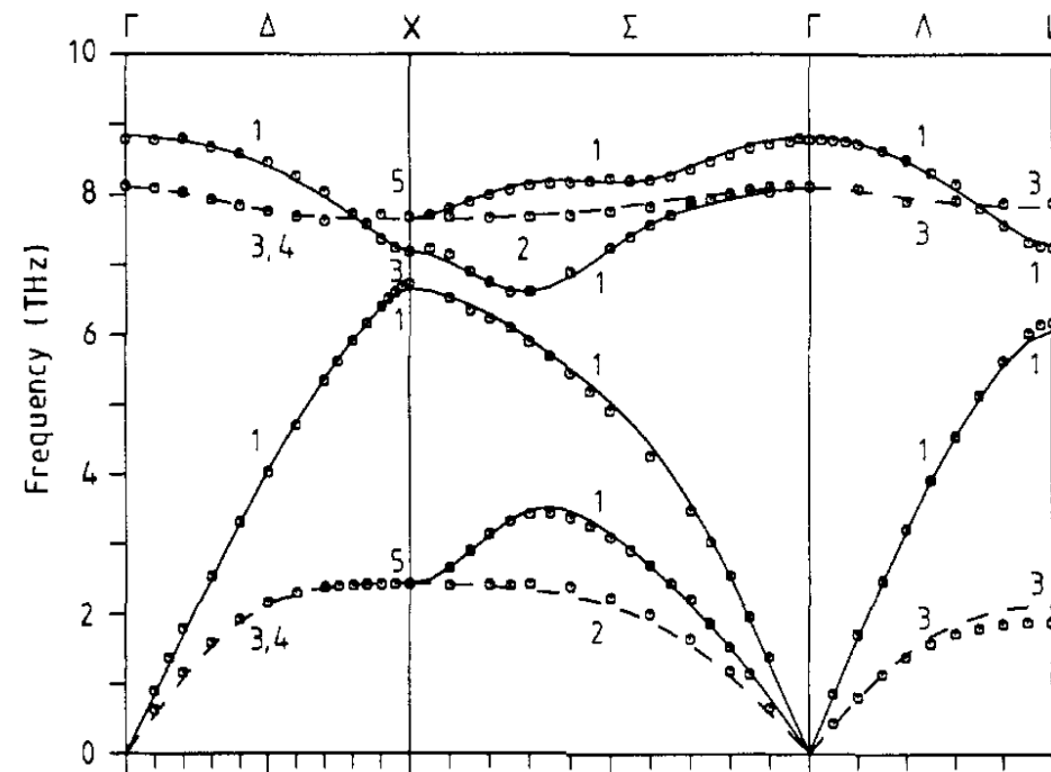
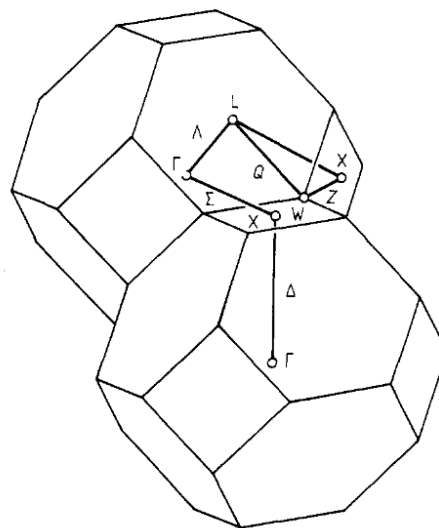
Acoustical Mode $\rightarrow$ sound wave



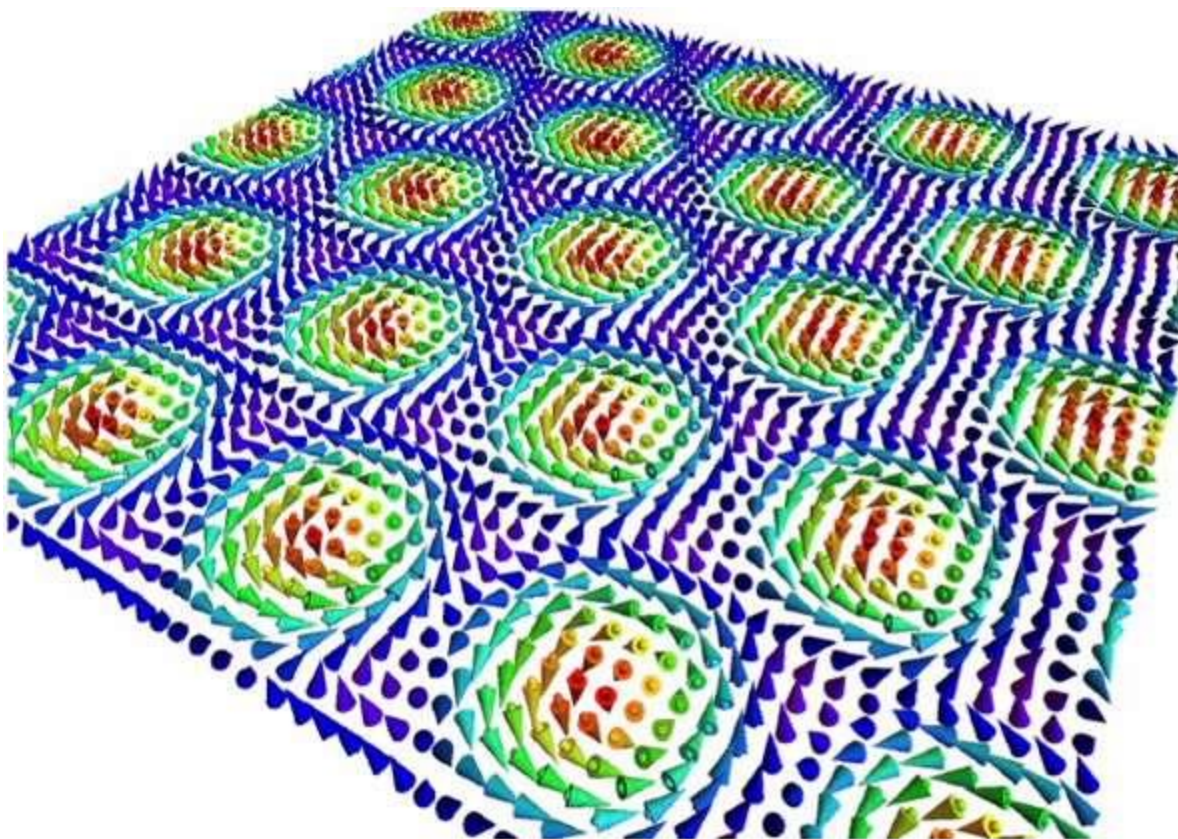
Optical Mode $\rightarrow$ interacts with light



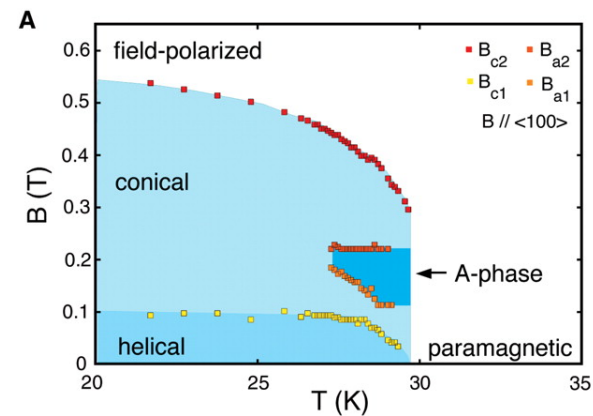
Zinc-blende structure



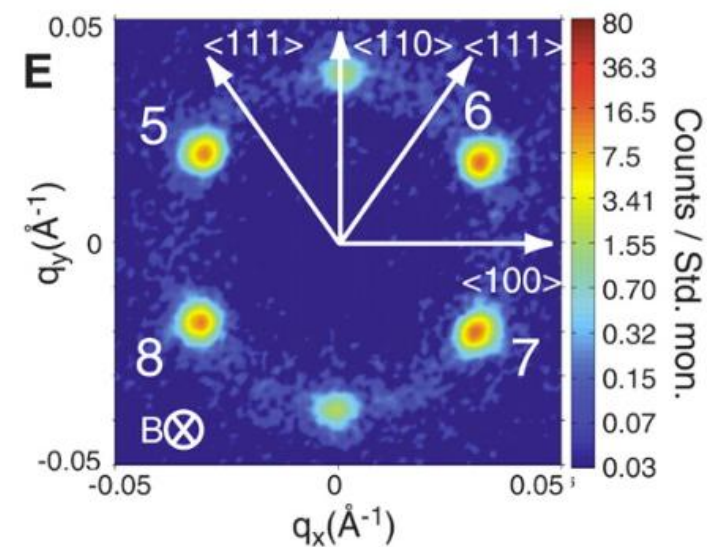
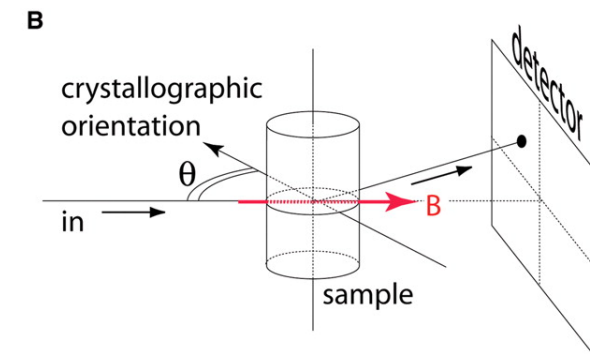
Skyrmion lattice

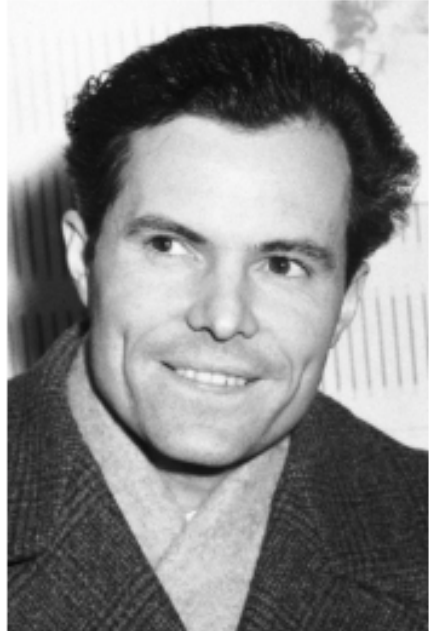


Prof. Christian Pfleiderer



Science 323, 915-919 (2009)





Rudolf Ludwig Mössbauer
Nobel Prize in Physics 1961

Born: 31 January 1929, Munich, Germany

Died: 14 September 2011

Affiliation at the time of the award: Technical University, Munich, Germany; California Institute of Technology (Caltech), Pasadena, CA, USA

Prize motivation: “for his researches concerning the resonance absorption of gamma radiation and his discovery in this connection of the effect which bears his name”

Prize share: 1/2

Photo from the Nobel Foundation archive.

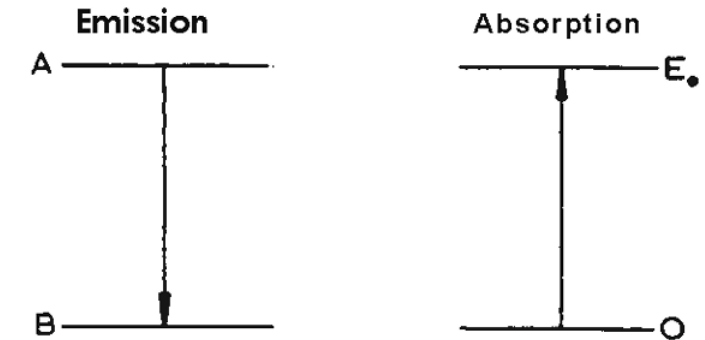


Fig. 1. Scheme of resonance absorption.

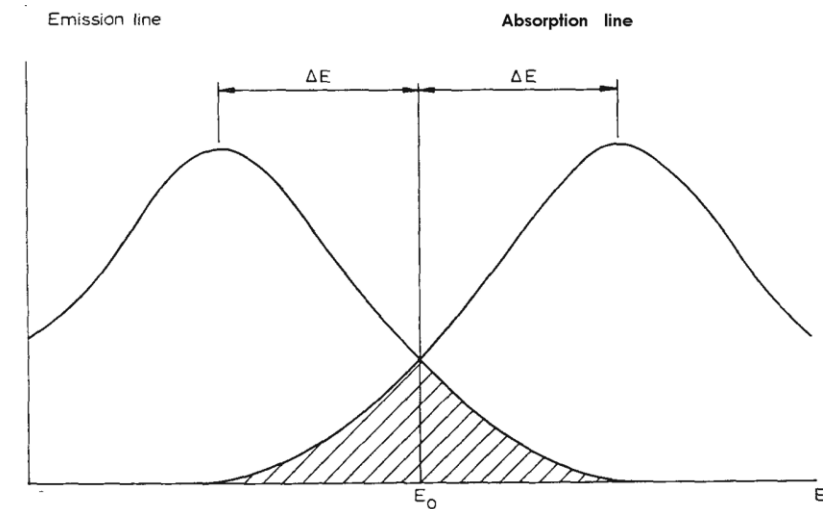


Fig. 2. Position and shape of the emission and absorption lines of a free nucleus (shown in the case of the 129-keV transition in ^{191}Ir for $T = 300^\circ\text{K}$).

For the standard 14.4 keV γ -ray of ^{57}Fe :

$$E_\gamma = 14.4 \times 10^3 \text{ eV}$$

$$M \approx 57 \text{ u} = 57 \times 931.5 \text{ MeV}/c^2$$

The recoil energy is:

$$E_{\text{recoil}} = \frac{p_\gamma^2}{2M} = \frac{E_\gamma^2}{2Mc^2}. \quad E_{\text{recoil}} \approx 2.0 \times 10^{-3} \text{ eV} \approx 2 \text{ meV}.$$

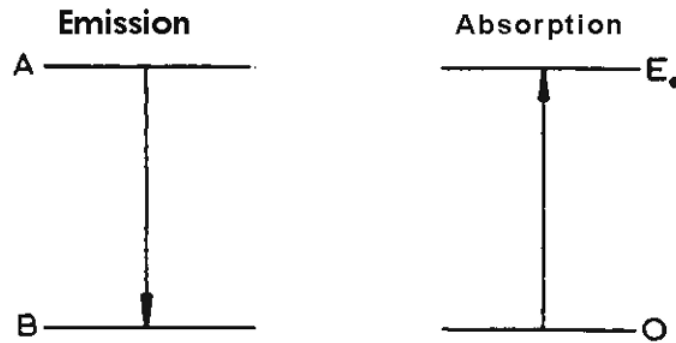


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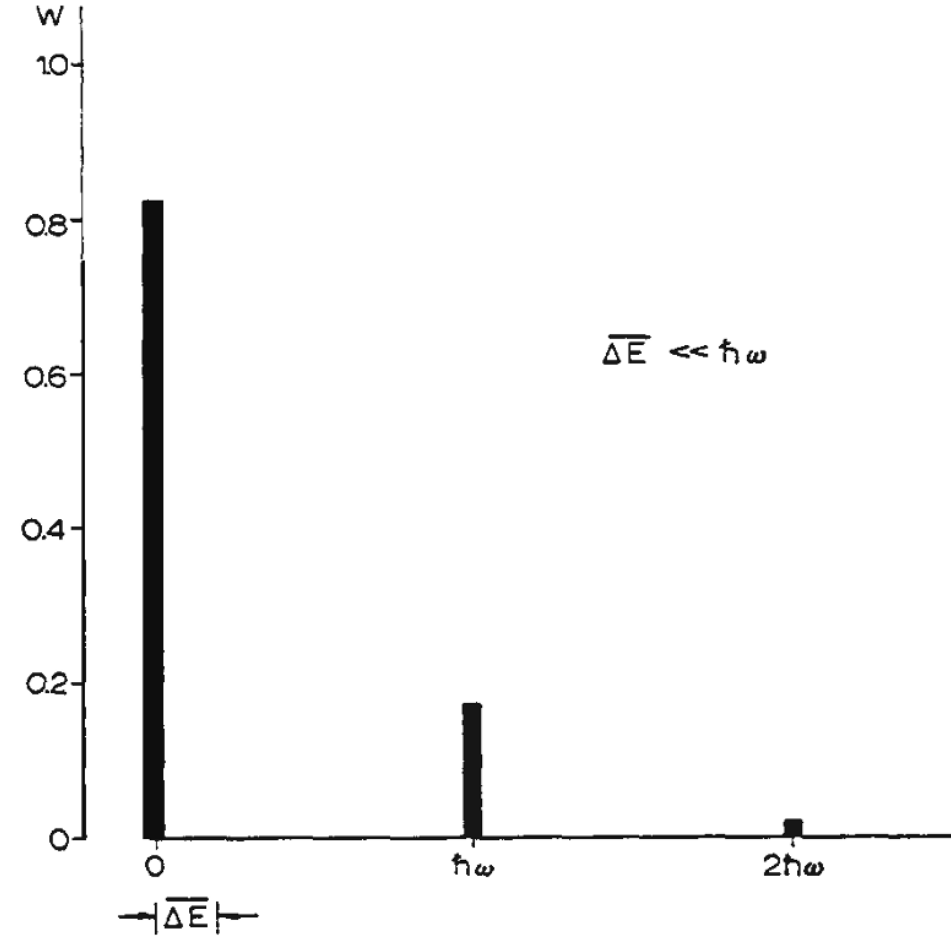
The linewidth of the transition is

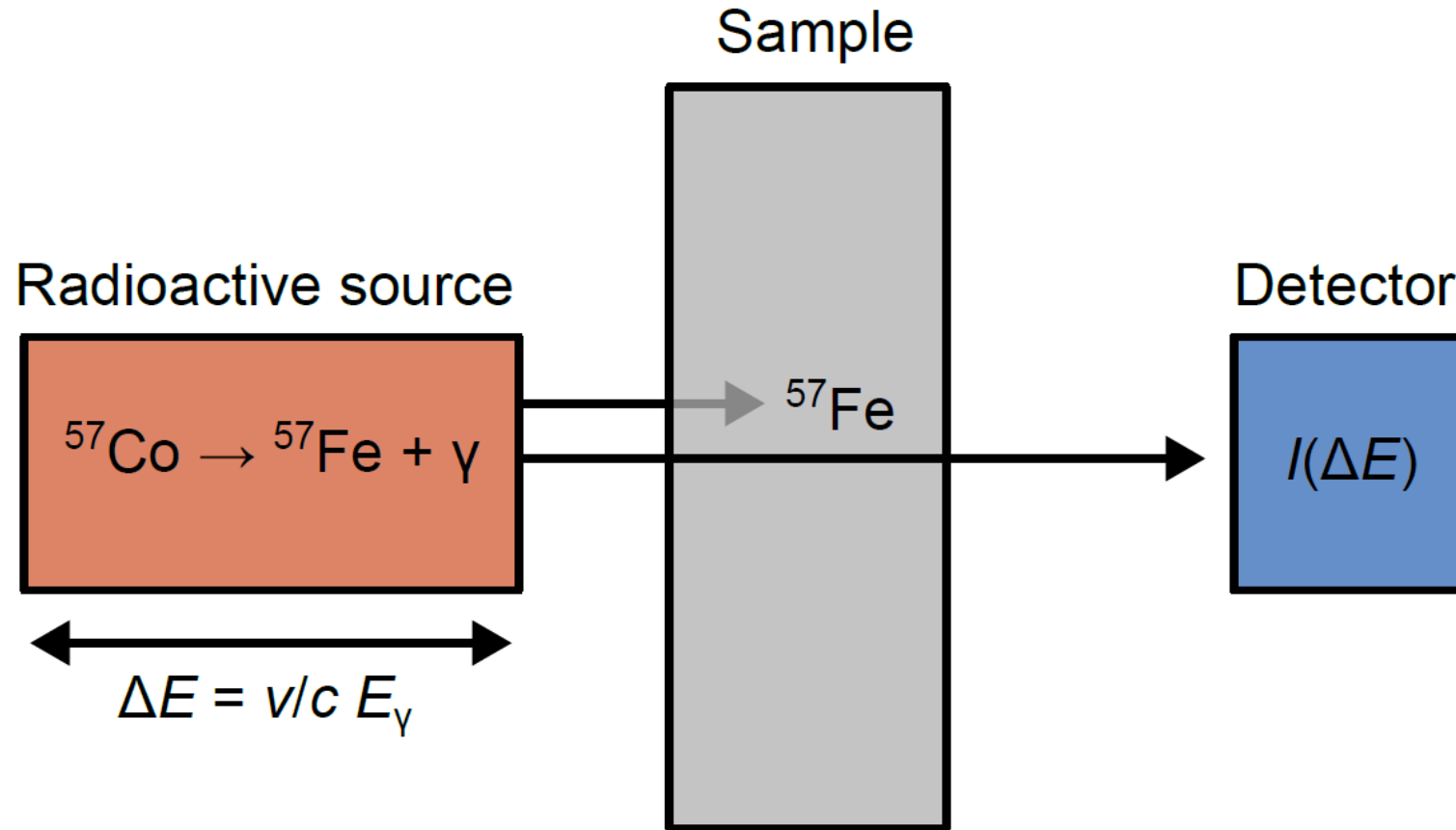
$$\Gamma \sim 4.7 \times 10^{-9} \text{ eV}$$

- In a crystal the mass M becomes very large (collective excitation of lattice vibrations)
→ recoil energy becomes negligible

$$E_{\text{recoil}} = \frac{p_{\gamma}^2}{2M} = \frac{E_{\gamma}^2}{2Mc^2}.$$

- More accurate picture:
 - the energy of the lattice vibrations is quantized (phonons)
 - at low enough temperature a significant fraction of the absorption processes do not generate a phonon excitation





The case of ^{57}Fe

